Semicompleteness of hypercube division sequences

A solution to MathDB 349741

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Abstract

For every positive dimension, the hypercube division sequence has a stronger property than semicompleteness: every integer between zero and the sum of any finite prefix is a sum of distinct terms of that prefix. The proof applies the classical complete-sequence criterion to an exact binomial identity. This proves Hoelscher's Conjecture 1.3 in the published paper (Conjecture 1.0.3 in its arXiv version).

Task 1: the original problem and its full solution

For integers $M, N \geq 1$, let C_M^N denote the maximum number of pieces obtained by cutting an M-dimensional hypercube with $N-1$ hyperplanes. For a fixed M, Hoelscher asks whether, for every $j > 2$ and every positive integer

$$k < \left\lfloor \frac{1}{j} \sum_{i=1}^{j} C_M^i \right\rfloor,$$

one can express k as a sum of distinct fractions chosen from $C_M^1/j, \ldots, C_M^j/j$. The cases $M = 1, 2, 3, 4$ are already proved in [1, Theorem 1.2 and Section 3]. We prove the assertion for all M. The subset-sum criterion used below is classical, due to Brown [2], and is also used in [1, Section 3]. We include its elementary proof; neither that criterion nor the region-count formula is claimed as new.

Theorem 1. *Fix $M \geq 1$ and $j \geq 1$, and put $S_{M,j} = \sum_{i=1}^{j} C_M^i$. For every integer t with $0 \leq t \leq S_{M,j}$ there is a subset $I \subseteq \{1, \ldots, j\}$ such that*

$$t = \sum_{i \in I} C_M^i.$$

Consequently $\{C_M^N\}_{N \geq 1}$ is semicomplete for every $M \geq 1$.

Lemma 1 (Finite-prefix form of the complete-sequence criterion)**.** *Let $a_1, \ldots, a_j$ be positive integers, with $a_1 = 1$. Suppose*

$$a_{n+1} \leq 1 + \sum_{i=1}^{n} a_i \qquad (1 \leq n < j).$$

Then every integer from 0 to $\sum_{i=1}^{j} a_i$ is the sum of a subset of the indexed terms.

Proof. For one term the possible sums are $0, 1$. Suppose the first n terms give every integer in $[0, S]$, where $S = \sum_{i=1}^{n} a_i$. Subsets omitting or including a_{n+1} give, respectively, all integers in $[0, S]$ and $[a_{n+1}, a_{n+1} + S]$. The hypothesis says that these two integer intervals overlap or are adjacent. Their union is every integer in $[0, S + a_{n+1}]$, proving the induction. □

Proof of Theorem 1. Use $\binom{n}{r} = 0$ for $r > n$. The standard hyperplane-region formula is

$$C_M^N = \sum_{r=0}^{M} \binom{N-1}{r}. \tag{1}$$

For completeness, the maximal number $R_M(n)$ of regions of n hyperplanes in $\mathbb{R}^M$ satisfies $R_M(n) = R_M(n-1) + R_{M-1}(n-1)$, with $R_M(0) = R_0(n) = 1$: the new hyperplane adds as many regions as the induced arrangement on it. General position attains the upper bound. Pascal's identity then gives $R_M(n) = \sum_{r=0}^{M} \binom{n}{r}$. The same maximum is attained in a hypercube: choose an interior point of each of the finitely many full-space regions and scale and translate the entire arrangement so all these points lie inside the hypercube. Convexity ensures that intersecting a region with the cube never creates more than one piece. This proves (1).

The hockey-stick identity gives, for every $N \geq 1$,

$$S_{M,N} = \sum_{r=0}^{M} \sum_{i=1}^{N} \binom{i-1}{r} = \sum_{r=0}^{M} \binom{N}{r+1}. \tag{2}$$

Subtracting (1), with $N+1$ in place of N, yields the exact identity

$$\boxed{1 + S_{M,N} - C_M^{N+1} = \binom{N}{M+1} \geq 0.} \tag{3}$$

Also $C_M^1 = 1$, so the lemma proves the asserted finite-prefix representation. The terms are positive and strictly increasing, since

$$C_M^{N+1} - C_M^N = \sum_{r=0}^{M-1} \binom{N-1}{r} \geq 1.$$

Thus distinct indices give distinct numerators, as required in the source.

Finally, if $j > 2$ and $0 < k < \lfloor S_{M,j}/j \rfloor$, then $0 < kj < S_{M,j}$. Apply the proved representation to $t = kj$ and divide by j. This gives precisely the required sum of distinct fractions. In fact the representation holds for every integer $0 \leq k \leq \lfloor S_{M,j}/j \rfloor$. □

Constructive interpretation. Consider the terms in descending order, selecting a term exactly when it does not exceed the current remainder. Suppose the remainder t is at most $S_{M,n}$. At $a_n = C_M^n$, either subtract a_n or leave t unchanged. The new remainder is at most $S_{M,n-1}$, because

$$t \geq a_n \implies t - a_n \leq S_{M,n-1}, \qquad t < a_n \implies t \leq a_n - 1 \leq S_{M,n-1}.$$

The final remainder is zero. No optimality in the number of selected terms is asserted.

Tasks 2–5: proposed follow-up directions

These are proposed questions, not additional solved claims.

Task 2. For a fixed M, classify the indices r for which deleting the term C_M^r and reindexing the remaining sequence preserves semicompleteness.

Task 3. Let $a_N = \sum_{r=0}^{d} b_r \binom{N-1}{r}$, where b_r are nonnegative integers, $b_0 \geq 1$, and $b_1 \geq 1$. Characterize the coefficient vectors for which (a_N) is semicomplete, and distinguish them from those for which every finite prefix has all subset sums between zero and its total.

Task 4. For fixed M, obtain a uniform asymptotic formula in the central range for the number of subsets $I \subseteq \{1, \ldots, j\}$ with $\sum_{i \in I} C_M^i = t$, as $j \to \infty$.

Task 5. For fixed M, determine the largest possible ratio, over positive representable targets in a prefix, between the number of terms selected by the descending algorithm above and the minimum number of distinct terms needed for that target.

Verification and status. The accompanying exact verifier checks the region recurrence against the closed form, enumerates finite subset sums, and reconstructs representations. These finite checks supplement the proof for all parameters. The published nine-page source and the current eight-page arXiv version were read in full; both retain the general conjecture. The dated literature audit reports bounded checks for earlier claims. All verification is by the authoring agent; no independent peer review is claimed.

References

[1] Z. Hoelscher, *Semicomplete Arithmetic Sequences, Division of Hypercubes, and the Pell Constant*, The PUMP Journal of Undergraduate Research **4** (2021), 108–116. doi:10.46787/pump.v4i0.2524; arXiv:2102.07083v1.

[2] J. L. Brown, Jr., *Note on Complete Sequences of Integers*, American Mathematical Monthly **68** (1961), 557–560. doi:10.2307/2311150.